POUGHKEEPSIE
Department 539
South Road Laboratory

January 9, 1959

FILE MEMO

SUBJECT: Control Codes for the 7030 System

BY: A. Padegs

This memo presents a comprehensive set of control codes for the 7030 System. The set contains 16 distinct function codes plus a No Operation code. Each code is represented by means of an 8-bit byte. The size and the general form of the control code byte has been chosen so that it be compatible with the 7030 System character codes.

The following factors have been taken into consideration in assigning the control codes within the set of 16 codes:

1. Simplicity of decoding. This applies primarily to the limited number of codes which have to be decoded in the channel adapter.

2. Assigning fixed codes to all functions which are common to more than one unit. In these cases, a code initiates the same function regardless to which unit the instruction is addressed, provided the addressed unit is designed to execute the function.

3. Assigning to logically related functions codes which differ by one bit, e.g.,

Backspace Block	Space Block
Backspace File	Space File

All other functions which are peculiar to individual units are assigned the remaining free codes. These codes usually have different meanings, depending upon the unit to which the instruction is addressed.

The control functions may be initiated in one of two ways:

1. By means of CONTROL instructions with the control code contained in the low order byte of the right effective address (bits 42 through 49). This method applies to the majority of input-output units.

H. G. Kolsky

2. By means of a WRITE instruction in which case the control code is transmitted as a data byte. The unit then has the task of distinguishing control codes from data to be written. This method of initiating control functions will be set up only for those units which need it, primarily the typewriter at this point.

In the case of CONTROL instructions, all 8-bit bytes consisting of bit combinations not listed in the attached code table will cause the EPK, Exchange Program Check, indication. Bits 32 through 41 in the right effective address of a CONTROL instruction will be ignored and will not cause any response from the unit.

When a WRITE instruction is used to transmit a control code to a console typewriter (or inquiry station), an Exchange Program Check is caused only when the high order bit of the byte is not zero (all information sent to the console by means of a WRITE instruction must have a zero high order bit).

The attached chart shows the code assignment for the console typewriter, console, 729 Tape and Tractor Tape. The 729 Tape has two sets of control codes, depending upon whether the tapes use a special 7030 adapter with a byte converter or not. The expanded set corresponds to the control functions presently specified in the manual. Only one of the two sets will be used. The capitalized letters underneath the name of the unit indicate the instruction by means of which the control function is initiated - WRITE or CONTROL.

The following decimal system is used to denote the 8-bit code:

code bit	0123 4567
decimal representation	XX - YY

The decimal numbers XX and YY stand for the value of the corresponding 4-bit binary quantity and are in the range 0 through 15, e.g.,

0101 1110 is represented as 5-14 .

AP/pkb

Attach.

cc: J. D. Calvert
T. G. Floros
R. A. Skov
H. K. Wild
7000 Product Planning
7000 Planning
J. W. Smith

CONTROL CODE CHART

Control Codes	Inquiry Station Console Typewriter (WRITE)	Console (CONTROL)	729 Tape without byte converter (CONTROL)	729 Tape with byte converter (CONTROL)	Tractor Tape (CONTROL)
0-0	Blank	No Operation	No Operation	No Operation	No Operation
0-14	Blank	Reserved Light Off	Reserved Light Off	Reserved Light Off	Reserved Light Off
0-15	End	Reserved Light On	Reserved Light On	Reserved Light On	Reserved Light On
1-14	Blank	Reserved Light Off	Reserved Light Off	High Density Mode	-- EPK
1-15	End	Reserved Light On	Reserved Light On	Low Density Mode	Load Cartridge
2-14	Blank	Reserved Light Off	Reserved Light Off	Erase Long Gap	-- EPK
2-15	End	Reserved Light On	Reserved Light On	ECC Mode	-- EPK
3-14	Blank	Reserved Light Off	Reserved Light Off	Space Block	Space Block
3-15	End	Reserved Light On	Reserved Light On	Space File	Space File
4-14	LF	Check Light On	Check Light On	Check Light On	Check Light On
4-15	CRLF	No Operation	Write Tape Mark	Write Tape Mark	Write Tape Mark
5-14	TAB	No Operation	Rewind	Rewind	Rewind
5-15	Blank	No Operation	Rewind and Unload	Rewind and Unload	Unload Cartridge
6-14	Black	No Operation	Even Parity	Even Parity and No-ECC Mode	File Protect Off
6-15	Red	No Operation	Odd Parity	Odd Parity and No-ECC Mode	File Protect On
7-14	BSP	No Operation	Backspace Block	Backspace Block	Backspace Block
7-15	Blank	Sound Gong	Erase Long Gap	Backspace File	Backspace File

AP/pkb 1/9/59